

16 Discrete Time Fourier Transform

In CT, for periodic $x(t)$, you first need a_k (The Fourier Series coefficient).

In DT, it will be the same - you need the Fourier coefficients for the transform.

Lets assume some periodic $x[n]$

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n}$$

(this is just the Fourier series).

If $x[n]$ is periodic and has a Fourier series representation in terms of a_k , then

$$X(e^{j\hat{\omega}}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\hat{\omega} - \frac{2\pi}{N}\right)$$

where $x[n]$ has period N .

This is an impulse train repeating forever.

Recall

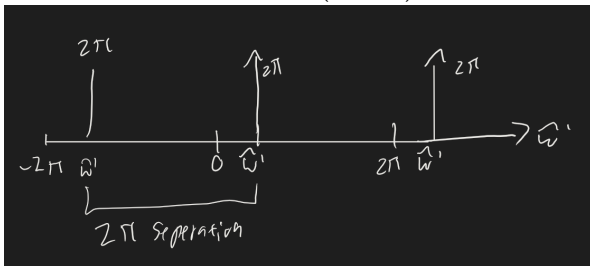
$$e^{j\hat{\omega}' n} \leftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \delta(\hat{\omega} - \hat{\omega}' - 2\pi l)$$

What does this look like?

For $l = 0$, we'll have $0 * 2\pi \delta(\hat{\omega} - \hat{\omega}')$

For $l = 1$, we'll have $1 * 2\pi \delta(\hat{\omega} - \hat{\omega}')$

For $l = -1$, we'll have $-2\pi \delta(\hat{\omega} - \hat{\omega}')$



How is this useful?

Example

Take $x[n] = \cos\left(\frac{\pi}{2}n\right)$, so $\hat{\omega}_0 = \frac{\pi}{2}$, $N = 4$.

We can write out Euler's identity

$$x[n] = \frac{1}{2} e^{j(1)\frac{\pi}{2}n} + \frac{1}{2} e^{j(-1)\frac{\pi}{2}n}$$

This gives us $a_1 = a_{-1} = \frac{1}{2}$.

The Fourier series expansion involves terms of the form $e^{j\hat{\omega}n}$.

for each $e^{j\text{stuff}}$. For the e^1 term only, we can make an $X(j\omega)$ - we have π high spikes at $-\frac{3\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}, \dots$

For the e^{-1} term, we have π high spikes at $-\frac{5\pi}{2}, -\frac{\pi}{2}, \frac{3\pi}{2}, \dots$

The two of them combine by just putting together their individual frequency components.

This gives us the Fourier transform for the periodic function as

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\hat{\omega} - \frac{2\pi k}{N}\right)$$

Summary

If you have a periodic $x[n]$ and want the DTFT,

1. Take the Fourier series, find a_k
2. To make it "math":

$$x(e^{j\hat{\omega}}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\hat{\omega} - \frac{2\pi}{N}k\right)$$

To make it "plot" (be lazy):

Multiply the x axis by $\frac{2\pi}{N}$ (or $\hat{\omega}_0$), and change the label to $\hat{\omega}$.

Turn the stems into arrows (continuous)

Multiply everything by 2π .

Properties of the Discrete Time Fourier Transform

1. Linearity

$$\begin{aligned} \text{if } x[n] &\leftrightarrow X(e^{j\omega}) \\ y[n] &\leftrightarrow Y(e^{j\omega}) \\ ax[n] + by[n] &= aX(e^{j\omega}) + bY(e^{j\omega}) \end{aligned}$$

2. Time Shifting

$$\begin{aligned} x[n] &\leftrightarrow X(e^{j\hat{\omega}}) \\ x[n - n_0] &\leftrightarrow e^{-j\hat{\omega}n_0} X(e^{j\hat{\omega}}) \end{aligned}$$

3. Convolutions and Multiplication

What about

$$x[n] \rightarrow (H[n]) \rightarrow y[n] = x[n] * h[n]$$

The DTFT gets

$$X(e^{j\hat{\omega}}) \rightarrow H(e^{j\hat{\omega}}) \rightarrow Y(e^{j\hat{\omega}})$$

What is $Y(e^{j\hat{\omega}})$?

$$Y(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} y[n] e^{-j\hat{\omega}n} = \sum_{n=-\infty}^{\infty} (x[n] * h[n]) e^{-j\hat{\omega}n}$$

$$Y(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k] h[n-k] \right) e^{-j\hat{\omega}n}$$

We can switch the order of the sum

$$= \sum_{k=-\infty}^{\infty} x[k] \left(\sum_{n=-\infty}^{\infty} h[n-k] e^{-j\hat{\omega}n} \right)$$

Remember:

$$\begin{aligned} h[n] &\leftrightarrow H(e^{j\omega}) \\ h[n-k] &\leftrightarrow e^{-j\hat{\omega}k} H(e^{j\hat{\omega}}) \end{aligned}$$

We get

$$\begin{aligned}
 Y(e^{j\hat{\omega}}) &= \sum_{k=-\infty}^{\infty} x[k]e^{-j\hat{\omega}k}H(e^{j\omega}) \\
 &= H(e^{j\hat{\omega}}) \sum_{k=-\infty}^{\infty} x[k]e^{-j\hat{\omega}k} \\
 Y(e^{j\hat{\omega}}) &= H(e^{j\hat{\omega}})X(e^{j\hat{\omega}})
 \end{aligned}$$

This is the same as in continuous time!

$$H(e^{j\hat{\omega}}) = \frac{Y(e^{j\hat{\omega}})}{X(e^{j\hat{\omega}})}$$

4. Multiplication Property

$$\begin{aligned}
 \text{if } x[n] &\leftrightarrow X(e^{j\hat{\omega}}) \\
 y[n] &\leftrightarrow Y(e^{j\hat{\omega}})
 \end{aligned}$$

Then

$$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{\theta=-2\pi}^{2\pi} Y(e^{j\theta})X(e^{j(\hat{\omega}-\theta)})d\theta$$

So if we have

$$y[n] = \frac{1}{2\pi} \int_{2\pi} Y(e^{j\hat{\omega}})e^{j\hat{\omega}n}d\hat{\omega}$$

then we also have

$$y[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\hat{\omega}})H(e^{j\hat{\omega}})e^{j\hat{\omega}n}d\hat{\omega}$$

Example

Lets take a difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n]$$

The time shifting property takes

$$x[n-n_0] \leftrightarrow e^{-j\hat{\omega}n_0}X(e^{j\hat{\omega}})$$

Taking the DTFT of the difference equation:

$$Y(e^{j\hat{\omega}}) - \frac{1}{2}e^{-j\hat{\omega}(1)}Y(e^{j\hat{\omega}}) = X(e^{j\hat{\omega}})$$

Rearranging,

$$H(e^{j\hat{\omega}}) = \frac{Y(e^{j\hat{\omega}})}{X(e^{j\hat{\omega}})} = \frac{1}{1 - \frac{1}{2}e^{-j\hat{\omega}}}$$

Example

If we had an input $x[n] = \left(\frac{1}{6}\right)^n u[n]$

We know

$$Y(e^{j\hat{\omega}}) = X(e^{j\hat{\omega}})H(e^{j\hat{\omega}})$$

and we can find that $X(e^{j\hat{\omega}}) = \frac{1}{1 - \frac{1}{6}e^{-j\hat{\omega}}}$.

Therefore,

$$Y(e^{j\hat{\omega}}) = \frac{1}{1 - \frac{1}{2}e^{-j\hat{\omega}}} \frac{1}{(1 - \frac{1}{6}e^{-j\hat{\omega}})}$$

We have two approaches from here. We could either use the raw integral, or we could expand and use transform pairs. First the easy way (pairs):

We can do partial fraction expansion here,

$$Y(e^{j\hat{\omega}}) = \frac{\frac{3}{2}}{(1 - \frac{1}{2}e^{-j\hat{\omega}})} - \frac{\frac{1}{2}}{(1 - \frac{1}{6}e^{-j\hat{\omega}})}$$

This gives us

$$y[n] = \frac{3}{2} \left(\frac{1}{2} \right)^n u[n] - \frac{1}{2} \left(\frac{1}{6} \right)^n u[n]$$

OR do the actual integral (annoying but general)

$$y[n] = \frac{1}{2\pi} \int_{2\pi} Y(e^{j\hat{\omega}}) e^{j\hat{\omega}n} d\hat{\omega}$$

	Periodic	Aperiodic OR Periodic
$x[n]$	DT Fourier Series	DT Fourier Transform
$x(t)$	CT Fourier Series	CT Fourier Transform